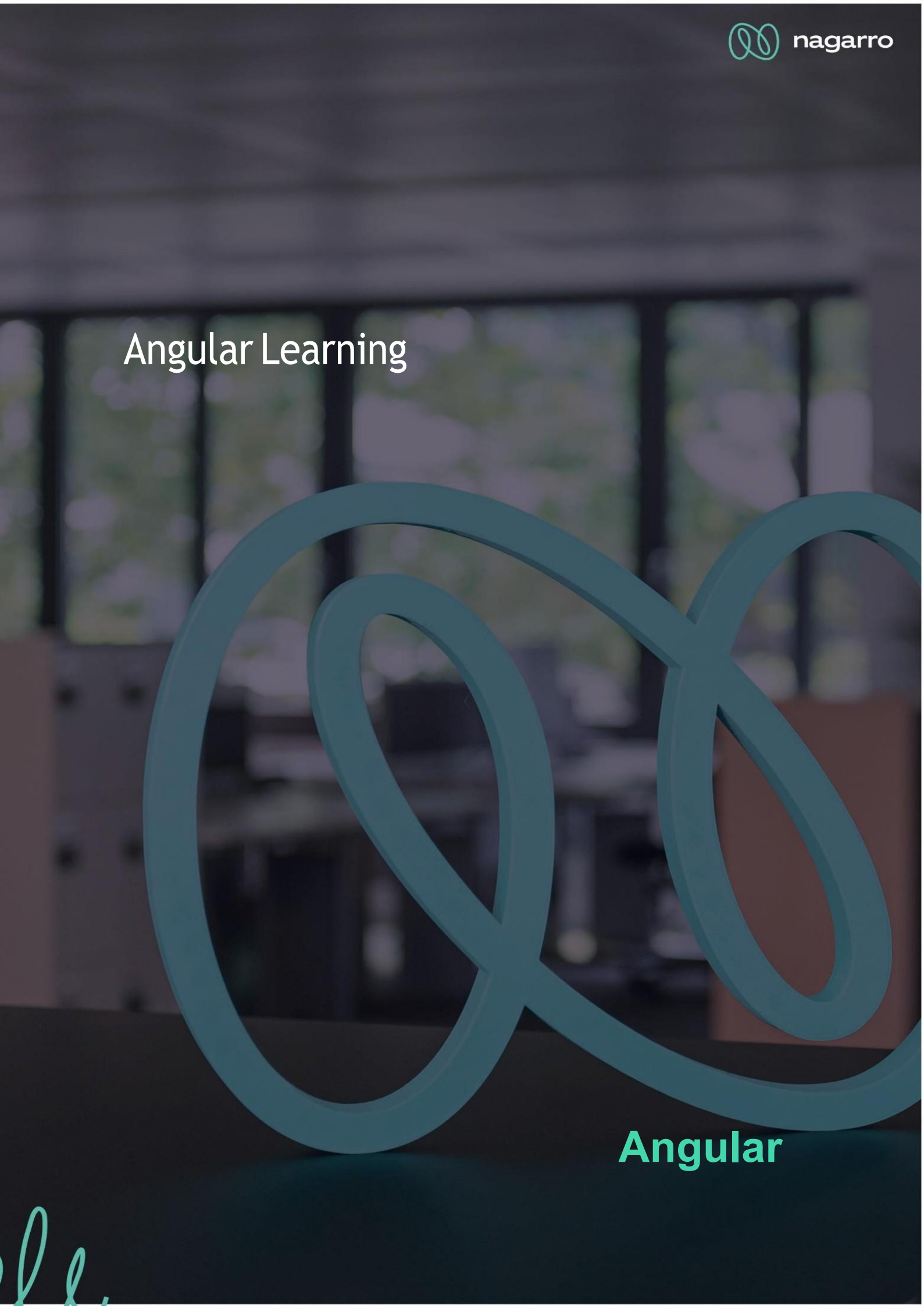
nagarro

Angular Learning



Angular

Overview

Topic	Sub-Topic	Duration (In Days)	Duration (In hours)	
Introduction	Overview of Angular, Installation, Setup & Project Structure	0.25	2	Total Duration: 2 Weeks
Components	Creating Components, Templates, Styles, Lifecycle Hooks	0.25	2	
Template syntax and data bindings	Property Binding, Event Binding, Two-way Binding	0.25	2	
Directives	Built-in Directives, Custom Directives	0.25	2	
Angular services & dependency injection	Creating Services, Dependency Injection	0.25	2	
Signals	Angular Signals	0.5	4	
Routing and navigation	Configuring Routes, Route Parameters, Lazy Loading, Router Guards	0.25	2	
Forms	Template-driven Forms, Reactive Forms, Form Validation	2	16	
Http client & APIs	Sending HTTP Requests, Fetching Data, Interceptors, Error Handling	0.5	4	
Testing	Unit Testing Components, Services, and Directives	0.25	2	
Deployment	Deploying to Server, GitHub, Firebase	0.25	2	
Assignment + Level Up		5	40	

Introduction

Angular is a powerful front-end web application framework developed by Google. It is widely used for building dynamic, single-page applications (SPAs). Angular provides a structured approach to application development, making it easier to manage large-scale projects with its robust set of tools and features.

This document will guide you through the basics of Angular, prerequisites for learning it, how to use it effectively, and what you can expect to learn from this guide.

Prerequisites

start using Angular, you need to set up your development environment. Here's a quick guide:

1. **Node.js and npm:** Angular requires Node.js and npm (Node Package Manager). Download and install them from nodejs.org.
2. **Angular CLI:** Install Angular CLI latest available version. Install it globally using npm:

   "npm install -g @angular/cli"

Learning Expectations

By the end of this guide, you should be able to:

1. **Understand the Angular Architecture:** Grasp the basic structure and concepts of Angular applications, including modules, components, services, and dependency injection.
2. **Create and Manage Angular Projects:** Use Angular CLI to scaffold and manage projects and understand the project structure.
3. **Develop Dynamic Web Applications:** Build SPAs with dynamic data binding, and routing.
4. **Implement Forms and Validation:** Create reactive and template-driven forms and apply validation.
5. **Consume RESTful APIs:** Integrate with backend services using Angular's HTTP Client.

Library Version

Please use the latest version of angular-cli.

Course Material

Topic	Documentation
Introduction	https://angular.dev/overview
Components	https://angular.dev/guide/components
Template syntax and data bindings	https://angular.dev/guide/templates
Directives	https://angular.dev/guide/directives
Angular services & dependency injection	https://angular.dev/guide/di
Signals	https://angular.dev/guide/signals
Routing and navigation	https://angular.dev/guide/routing
Forms	https://angular.dev/guide/forms
Http client & APIs	https://angular.dev/guide/http
Testing	https://angular.dev/guide/testing
Deployment	https://angular.dev/tools/cli/deployment

Learning Resources

- **Official Angular Documentation:** The official Angular documentation is comprehensive and regularly updated.
- **Tutorials:**
  - Foundational Concepts Playlist
  - Mastering Angular 16: Build Powerful Web Applications
  - Angular Crash Course 2024 (for Beginners)
  - Learn Angular A-Z: Complete Tutorial for Beginners
  - Angular V17
  - Complete Angular Tutorial For Beginners - Tektutorialshub
  - Angular Project ideas for practice - Top 28 Angular Projects to Boost Your Skills in 2025!
  - Debug Angular Apps Efficiently 🚀 Best Tools & Techniques!

Level Up

https://nagarro.sharepoint.com/sites/university/SitePages/LearningManagementSystem.aspx?courseId=8597

Additional Reference Links:

- **LinkedIn Learning**
  - Explore Web Development with Angular
  - Learning Angular
  - Angular Essential Training